

Solutions – Quiz 4

ECON 320 – Introduction to Mathematical Economics

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Part A – Multiple Choice Questions (1 mark each)

1. (a)

$$\frac{d}{dx} \ln(x^2 + 1) = \frac{1}{x^2 + 1} \cdot (2x) = \frac{2x}{x^2 + 1}.$$

2. (b)

$$\frac{d}{dx} e^{2x} = 2e^{2x}.$$

3. (b)

$$y = e^{3x-5} \Rightarrow \frac{dy}{dx} = 3e^{3x-5}.$$

4. (b) $\ln(x)$ is defined for $x > 0$.

5. (c)

$$f(x) = \ln(e^{2x}) = 2x \Rightarrow f'(x) = 2.$$

6. (b) For $y = ae^{bx}$,

$$E_{yx} = \frac{dy}{dx} \frac{x}{y} = abe^{bx} \frac{x}{ae^{bx}} = b.$$

7. (a) $\ln(x)$ is increasing ($f'(x) = 1/x > 0$) and concave ($f''(x) = -1/x^2 < 0$).

Part B – Analytical Problems (13 marks total)

Question 1 (4 marks)

Utility: $U(x) = \ln(x + 2)$.

(a) **Derivatives.**

$$U'(x) = \frac{1}{x+2}, \quad U''(x) = -\frac{1}{(x+2)^2}.$$

(b) **Monotonicity and Concavity.**

$$U'(x) > 0 \quad \forall x > -2 \Rightarrow \text{Utility increasing.}$$

$$U''(x) < 0 \quad \forall x > -2 \Rightarrow \text{Utility concave.}$$

(c) **Economic interpretation.** Concavity indicates **diminishing marginal utility of income**: each extra dollar adds less to utility.

Question 2 (4 marks)

Production: $Q(L) = 20L^{0.5}e^{-0.05L}$.

(a) **Derivative.**

$$Q'(L) = 20e^{-0.05L}(0.5L^{-0.5} - 0.05L^{0.5}) = 20e^{-0.05L}L^{-0.5}(0.5 - 0.05L).$$

(b) **FOC.**

$$0.5 - 0.05L = 0 \Rightarrow L^* = 10.$$

(c) **SOC.**

$$Q''(L) = 20e^{-0.05L}L^{-1.5}(-0.025L^2 + 0.0025L - 0.25),$$

which is negative at $L = 10$, so L^* yields a maximum.

(d) **Interpretation.** The exponential term models diminishing productivity—beyond L^* , crowding or inefficiency lowers output.

Question 3 (5 marks)

Demand: $Q(P) = 100e^{-0.2P}$. Revenue: $R(P) = PQ(P) = 100Pe^{-0.2P}$.

(a) **Derivative.**

$$R'(P) = 100e^{-0.2P}(1 - 0.2P).$$

(b) **FOC.**

$$1 - 0.2P = 0 \Rightarrow P^* = 5.$$

(c) **Maximum revenue.**

$$R(5) = 100(5)e^{-1} = 500/e \approx 183.94.$$

(d) **SOC.**

$$R''(P) = 100e^{-0.2P}(-0.2(1 - 0.2P) - 0.2) = 100e^{-0.2P}(-0.4 + 0.04P).$$

At $P = 5$: $R''(5) = 100e^{-1}(-0.2) < 0 \Rightarrow$ maximum.

(e) **Interpretation.** At low prices, higher prices increase total revenue; beyond $P = 5$, the exponential decay in demand dominates, reducing total revenue.